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24MA313

JSS MAHAVIDYAPEETHA

JSS SCIENCE AND TECHNOLOGY UNIVERSITY, MYSURU

Third Semester B.E. Degree: Test-I

Department of Mathematics

Engineering Mathematics-III

Duration: 1 Hour 15 Minutes Max. Marks: 30

Instructions:

1. Answer all questions in **PART-A**.
2. Questions in **PART-B** have internal choice.

PART-A

Q.No.	CO	CD	PI	Marks
1	CO-1	1,3	1.1.1	07
	Given the equation of the regression lines: $4x - 5y + 33 = 0$ and $20x - 9y = 107$. Find the mean of x and y and the correlation coefficient. Find σ_y if $\sigma_x^2 = 9$.			
2	CO-2	1,3	1.1.1	08
	An urn A contains 2 white and 4 black balls. A second urn contains 5 white and 7 black balls. A ball is transferred from A to B and then a ball is drawn from B. Find the probability that it is white?			

PART-B

3 (a)	CO-1	1,3	1.1.1	08
	Fit a second degree parabola of the form $y = ax^2 + bx + c$ by the method of least squares for the following data:			

x :	1	2	3	4	5	6	7	8	9
y :	2	6	7	8	10	11	11	10	9

3 (b)	CO-2	1,3	1.1.1	07
	If A, B and C are independent events, show that the events: (i) $\overline{A}$ and $\overline{B}$ are also independent. (ii) $\overline{A}$ and B are also independent. (iii) A and $\overline{B}$ are also independent.			
	OR			

4 (a)	CO-1	1,3	1.1.1	07
	Obtain the lines of regression and hence find the coefficient of correlation for the following data:			

x :	10	14	18	22	26	30
y :	18	12	24	6	30	36

4 (b)	CO-2	1,3	1.1.1	08
	Three students X Y and Z write an examination. Their chances of passing are $\frac{1}{2}$, $\frac{1}{3}$ and $\frac{1}{4}$ respectively. Find the probability that (i) all of them pass (ii) at least one of them pass (iii) at least two of them pass.			

Course Outcome: At the end of the course the students will have the ability to	
CO-1:	Apply least square method to fit a curve for the given data and evaluate the correlation coefficient and regression lines for the data.
CO-2:	Determine the nature of the events and hence calculate the appropriate probabilities of the events.

Performance Indicators:

1.1.1	Apply mathematical techniques such as calculus, linear algebra, and statistics to solve problems.
1.3.1	Apply fundamental engineering concepts to solve engineering problems.

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